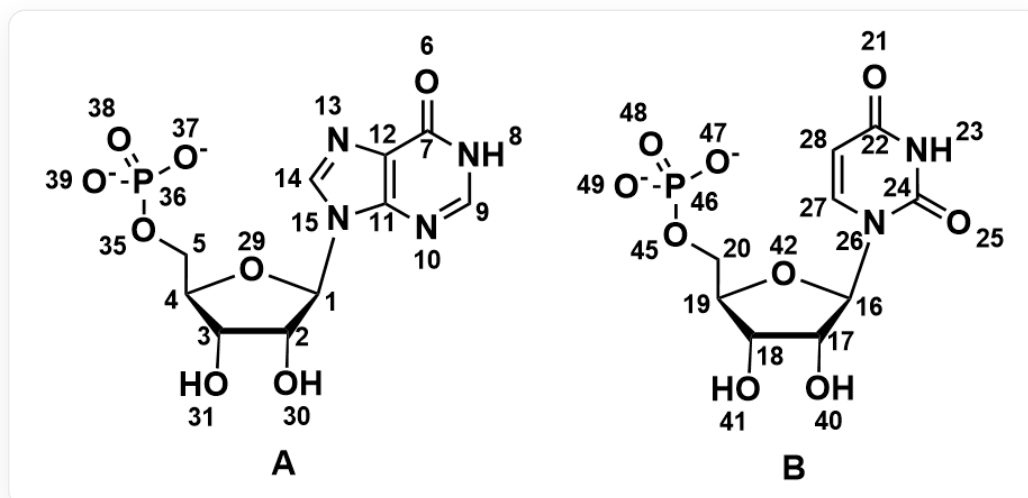
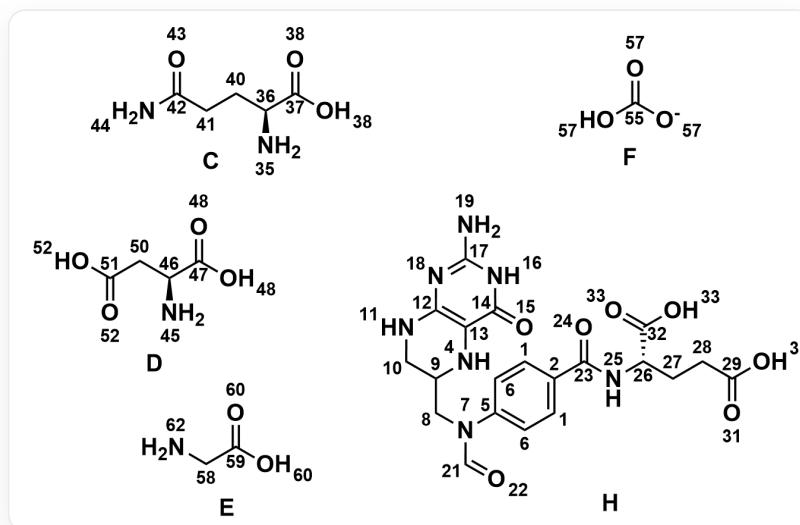


Question



In the figure, the SMILES of **A**: [O:6]=[C:7]1[C:12]([N:13]=[CH:14][N:15]2[C@H:1]3[C@H:2]([OH:30])[C@H:3]([OH:31])[C@@H:4]([CH2:5][O:35][P:36]([O-:37])([O-:38])=[O:39])[O:29]3)=[C:11]2[N:10]=[CH:9][NH:8]1; the SMILES of **B**: [OH:40][C@@H:17]1[C@H:18]([OH:41])[C@@H:19]([C:20][O:45][P:46]([O-:47])([O-:49])=[O:48])[O:42][C@H:16]1[N:26]2[C:24]([O:25])[NH:23][C:22]([O:21])[C:28]=[C:27]2

The above two nucleotides **A** (1) and **B** (2) can be synthesized in organisms from the following species **C** ~ **H**:



In the figure, the SMILES of **C**: [NH2:35][C@@H:36]([CH2:40][CH2:41][C:42]([O:43])[NH2:44])[C:37]([O:38])[OH:38]; the SMILES of **D**: [NH2:45][C@@H:46]([CH2:50][C:51]([OH:52])=[O:52])[C:47]([OH:48])=[O:48]; the SMILES of **E**: [O:60]=[C:59]([OH:60])[CH2:58][NH2:62]; the SMILES of **F**: [OH:57][C:55]([O-:57])=[O:57]; the SMILES of **H**: [O:15]=[C:14]([NH:16][C:17]([NH2:19])=[N:18]1[C:13]([NH:4]2)=[C:12]1[NH:11][CH2:10][CH:9]2[CH2:8][N:7]([CH:21]=[O:22])[C:5]3=[C:6][C:1]=[C:2]([C:23]([NH:25][C@H:26]([C:32]([OH:33])=[O:33])[CH2:27][CH2:28][C:29]([OH:31])=[O:31])=[O:24])[C:1]=[C:6]3

The same numbering is used for atoms in the same chemical environment in $\mathbf{C} \sim \mathbf{H}$ to facilitate atomic correspondence.

Based on the nucleotide metabolic pathway, please find the correspondence between the atoms of the base in the nucleotide and the atoms of the raw material species in the organism, recorded as:

$$(b_i, a_i)$$

where b_i refers to the atom number in the nucleotide in the correspondence, and a_i refers to the number of the raw material atom corresponding to the base atom in the raw material species synthesized in the organism. Different b_i may correspond to the same a_i .

For the correspondence in each nucleotide j , define the calculation method of x_j as:

$$x_j = \sum_i (a_i \times b_i), \quad a_i \in j$$

Please calculate the value of $\frac{x_1}{x_2}$, keep four significant figures, and select the correct result from the following options, requiring the deviation between the calculated result and the option to be within 1%, otherwise select option A: All other options are incorrect.

A. All other options are incorrect

B. 2.029

C. 0.6164

D. 0.2729

E. 0.4434

F. 0.4928

G. 0.5357

H. 1.165

I. 0.4711

J. 1.000

K. 0.7500

Answer

Correct Answer: **F**

Detailed Explanation

To solve this problem, we first need to identify the types of nucleotides. Nucleotide **A** is inosine monophosphate (inosinic acid) IMP, and nucleotide **B** is uridine monophosphate UMP.

CHECKPOINT

2 PTS

Nucleotide **A** is inosine monophosphate (inosinic acid) IMP, and nucleotide **B** is uridine monophosphate UMP

Secondly, we need to determine the types of each raw material: **C** is glutamine, **D** is aspartic acid, **E** is glycine, **F** is bicarbonate, and **H** is formyltetrahydrofolate.

CHECKPOINT

3 PTS

C is glutamine, **D** is aspartic acid, **E** is glycine, **F** is bicarbonate, and **H** is formyltetrahydrofolate

Next, find the correspondence between the atoms of the bases in the nucleotides and the atoms of the raw material species in the organism.

For IMP/UMP, we can first correlate the traditional numbering method with the atom numbering given in the question. According to the traditionally defined numbering, combined with the knowledge of nucleotide biosynthesis, find the corresponding atom number in the corresponding raw material, and the following table can be obtained:

Nucleotide	Atom Number in the Question	Traditional Atom Number	Raw Material	Corresponding Raw Material Atom Number
IMP	6	(Oxygen of carbon 6)	Oxygen in F	57
IMP	7	6	Carbon in F	55
IMP	8	1	Amino group in D	45
IMP	9	2	Carbon of the formyl group in H	21
IMP	10	3	Amide amino group in C	44
IMP	11	4	Carboxylic carbon in E	59
IMP	12	5	α - carbon of the carboxyl group in E	58
IMP	13	7	Amino group in E	62
IMP	14	8	Carbon of the formyl group in H	21
IMP	15	9	Amide amino group in C	44
UMP	21	(Oxygen of carbon 4)	Oxygen of the side-chain carboxyl group in D	52
UMP	22	4	Carbon of the side-chain carboxyl group in D	51
UMP	23	5	Amide amino group in C	44
UMP	24	6	Carbon in F	55
UMP	25	(Oxygen of carbon 6)	Oxygen in F	57
UMP	26	1	Amino group in D	45
UMP	27	2	Carbon connected to the amino group in D	46

Nucleotide	Atom Number in the Question	Traditional Atom Number	Raw Material	Corresponding Raw Material Atom Number
UMP	28	3	α - carbon of the side-chain carboxyl group in D	50

CHECKPOINT

3 PTS

Correspondence in IMP

$$(b_6, a_{57}) / (b_7, a_{55}) / (b_8, a_{45}) / (b_9, a_{21}) / (b_{10}, a_{44}) / (b_{11}, a_{59}) / (b_{12}, a_{58}) / (b_{12}, a_{58}) / (b_{13}, a_{62}) / (b_{14}, a_{21}) / (b_{15}, a_{44})$$

CHECKPOINT

3 PTS

Correspondence in UMP $(b_{21}, a_{52}) / (b_{22}, a_{51}) / (b_{23}, a_{44}) / (b_{24}, a_{55}) / (b_{25}, a_{57}) / (b_{26}, a_{45}) / (b_{27}, a_{46}) / (b_{28}, a_{50})$

Finally, according to the calculation method of x_j defined in the question, calculate the values of x_1 and x_2 corresponding to IMP and UMP respectively, and find the ratio between them.

First calculate x_1 (IMP), according to the formula $x_j = \sum_i (a_i \times b_i)$ and the IMP atom correspondence table, we have:

$$x_1 = (6 \times 57) + (7 \times 55) + (8 \times 45) + (9 \times 21) + (10 \times 44) + (11 \times 59) + (12 \times 58) + (13 \times 62) + (14 \times 21) + (15 \times 44)$$

Calculate the product of each term:

$$* 6 \times 57 = 342$$

$$* 7 \times 55 = 385$$

$$* 8 \times 45 = 360$$

$$* 9 \times 21 = 189$$

$$* 10 \times 44 = 440$$

$$* 11 \times 59 = 649$$

$$* 12 \times 58 = 696$$

$$* 13 \times 62 = 806$$

$$* 14 \times 21 = 294$$

$$* 15 \times 44 = 660$$

Add all the products together:

$$x_1 = 342 + 385 + 360 + 189 + 440 + 649 + 696 + 806 + 294 + 660 = 4821$$

CHECKPOINT

1 PTS

$x_1 = 4821$ corresponding to IMP

Then calculate x_2 (UMP), similarly, according to the UMP atom correspondence table:

$$x_2 = (21 \times 52) + (22 \times 51) + (23 \times 44) + (24 \times 55) + (25 \times 57) + (26 \times 45) + (27 \times 46) + (28 \times 50)$$

Calculate the product of each term:

$$* 21 \times 52 = 1092$$

$$* 22 \times 51 = 1122$$

$$* 23 \times 44 = 1012$$

$$* 24 \times 55 = 1320$$

$$* 25 \times 57 = 1425$$

$$* 26 \times 45 = 1170$$

$$* 27 \times 46 = 1242$$

$$* 28 \times 50 = 1400$$

Add all the products together:

$$x_2 = 1092 + 1122 + 1012 + 1320 + 1425 + 1170 + 1242 + 1400 = 9783$$

CHECKPOINT

1 PTS

 $x_2 = 9783$ corresponding to UMP

Calculate the value of $\frac{x_1}{x_2}$:

$$\frac{x_1}{x_2} = \frac{4821}{9783} \approx 0.49279 \approx 0.4928$$

CHECKPOINT

1 PTS

$$\frac{x_1}{x_2} = 0.4928$$